1. Basic Circle Algorithm

Definition: Draws a circle using the circle equation:

$$x^2 + y^2 = r^2$$

or

$$y = \sqrt{r^2 - x^2}$$

Steps:

1. Input center (x_c, y_c) and radius r.
2. Calculate y for each x.
3. Plot the eight symmetric points.
4. Repeat until $x > y$.

Advantages:

- Simple to understand.
- Uses the circle equation directly.

Disadvantages:

- Uses square root and floating-point calculations.
- Slow for graphics applications.

2. Midpoint Circle Algorithm

Definition: Uses a **decision parameter** to choose the next pixel without using square roots.

Initial Values:

- $x = 0$
- $y = r$
- $P = 1 - r$

Decision:

- If $P < 0$: choose $(x + 1, y)$
- Otherwise: choose $(x + 1, y - 1)$

Advantages:

- Fast.
- Uses only integer arithmetic.
- More efficient than the basic algorithm.

3. Bresenham's Circle Algorithm

Definition: An improved circle drawing algorithm that uses only integer calculations.

Initial Values:

- $x = 0$
- $y = r$
- $d = 3 - 2r$

Decision:

- If $d < 0$: choose $(x + 1, y)$
- Otherwise: choose $(x + 1, y - 1)$

Advantages:

- Very fast.
- No floating-point operations.
- Highly efficient for raster graphics.

Algorithm	Decision Parameter	Uses Square Root	Speed
Basic Circle Algorithm	No	Yes	Slow
Midpoint Circle Algorithm	$P = 1 - r$	No	Fast
Bresenham's Circle Algorithm	$d = 3 - 2r$	No	Very Fast